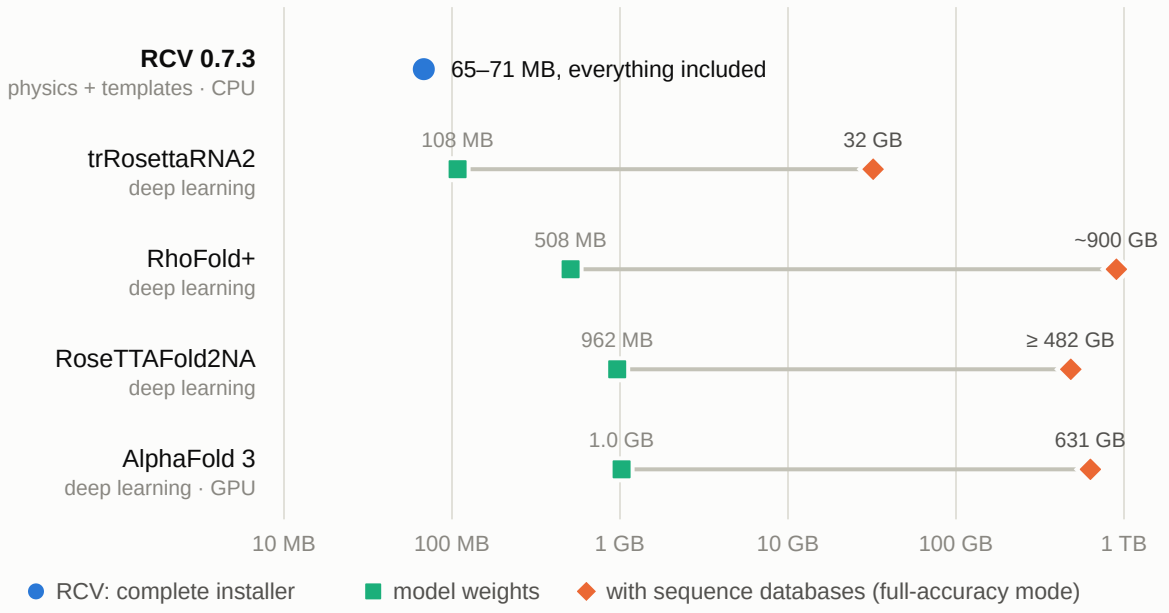
# RCV next to deep-learning RNA 3D structure predictors

Size, hardware and RNAs that move. RCV: physics + experimental templates, no deep learning, no GPU.

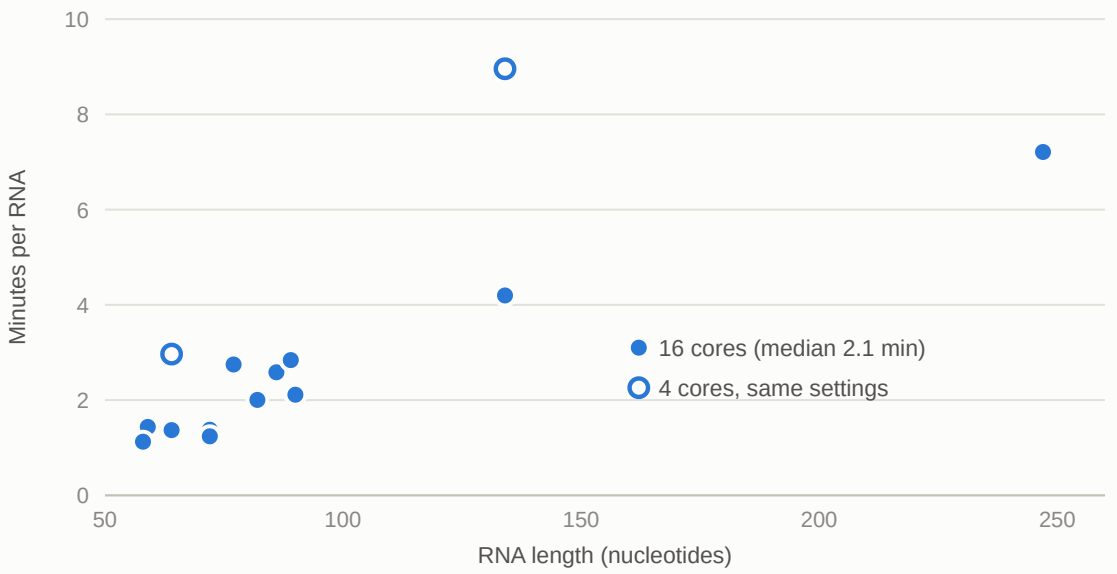
## a Download and disk space

Log scale. Other tools: Python/GPU software not counted.




## b Time per RNA on a laptop CPU, no GPU

RCV 0.7.3, 5 models from 40 trajectories, CASP16 targets. Intel Core Ultra 9 285H, 16 cores.

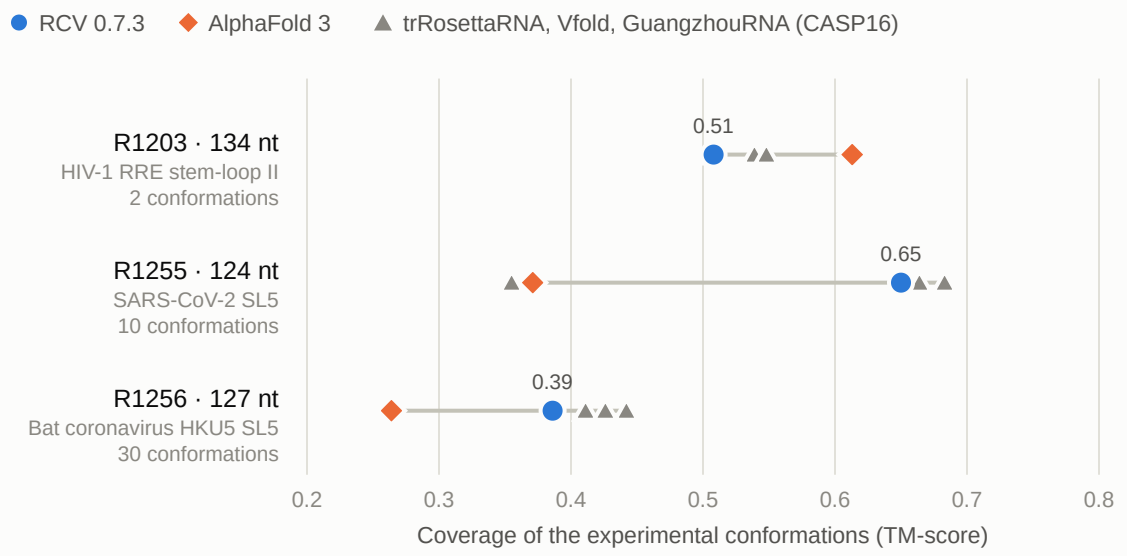



RCV peak memory: 1.3 GB (247 nt, all 16 workers together).

Hardware the others need (documentation):
AlphaFold 3 — NVIDIA GPU, compute capability ≥ 8.0 (A100/H100 tested), ≥ 64 GB RAM; genetic search on CPU.

## c RNAs solved in several conformations

CASP16: best TM-score of the 5 models for each experimental conformation, averaged.




## d For an RNA that moves

What each tool provides, from its documentation.

| | **RCV 0.7.3** | AlphaFold 3 | trRosetta-RNA2 | RhoFold+ | RoseTTA-Fold2NA |
|---|---|---|---|---|---|
| Conformations per run | ensemble[1] | 5 per seed[2] | 1[3] | 1 | 1 |
| Alternative 2D states, populations | ✓ | – | – | – | – |
| Temperature, salt, ligand | ✓ | – | – | – | – |
| Switching paths, barriers, rates | ✓ | – | – | – | – |
| Fit to SHAPE/DMS probing data | ✓ | – | – | – | – |
| Interactive ensemble viewer | ✓ | –[4] | – | – | – |
| Runs without a GPU | ✓ | – | not stated | ✓ | not stated |
| Sequence databases needed | none | 630 GB | 32 GB[5] | ~900 GB[5] | ≥ 481 GB |

[1] structurally distinct conformations from different 2D/template hypotheses (1–30, default 8)
[2] diffusion samples of one prediction; more seeds give more samples
[3] the trRosettaRNA2 paper describes conformers; the default output is one model
[4] AlphaFold 3 results open in RCV (0.7.3), so RCV's analyses apply to them
[5] optional: single-sequence mode without databases is less accurate

Sizes and requirements as documented by each tool (checked 7 Nov 2026); weight files measured on the download servers. Accuracy on the 15 CASP16 targets: benchmark/casp16/results.